



Git Commands You MUST Know!

Basic Setup

Command	Use Case
<code>git config --global user.name "Your Name"</code>	Set your Git username
<code>git config --global user.email "your@email.com"</code>	Set your Git email
<code>git config --list</code>	Verify your Git configuration
<code>git init</code>	Initialize a new Git repository
<code>git clone <repo-url></code>	Copy (clone) an existing repo from GitHub

Working with Files

Command	Use Case
<code>git status</code>	Check current file changes
<code>git add <file></code>	Stage a specific file for commit
<code>git add .</code>	Stage all modified files
<code>git rm <file></code>	Remove a file from the repo
<code>git mv <old> <new></code>	Rename or move a file

Committing Changes

Command	Use Case
<code>git commit -m "your message"</code>	Save staged changes with a message
<code>git commit -am "message"</code>	Add and commit all modified files at once
<code>git log</code>	View commit history
<code>git log --oneline</code>	Short and clean commit history
<code>git show <commit-id></code>	View details of a specific commit

Branching & Merging

Command	Use Case
<code>git branch</code>	List all branches
<code>git branch <branch-name></code>	Create a new branch
<code>git checkout <branch-name></code>	Switch to another branch
<code>git checkout -b <branch-name></code>	Create and switch to a new branch
<code>git merge <branch-name></code>	Merge a branch into the current one
<code>git branch -d <branch-name></code>	Delete a branch

Remote Repositories (GitHub)

Command	Use Case
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<code>git remote add origin <repo-url></code>	Connect local repo to GitHub
<code>git remote -v</code>	Verify remote connections
<code>git push -u origin main</code>	Push local commits to GitHub
<code>git pull origin main</code>	Pull the latest updates from GitHub
<code>git fetch</code>	Get latest commits without merging
<code>git push</code>	Upload your changes to GitHub

Undoing Changes

Command	Use Case
<code>git reset <file></code>	Unstage a file (keep changes)
<code>git reset --hard</code>	Discard all local changes
<code>git revert <commit-id></code>	Undo a commit safely
<code>git checkout -- <file></code>	Revert a file to last committed state

Advanced Useful Commands

Command	Use Case
<code>git stash</code>	Temporarily save uncommitted changes
<code>git stash pop</code>	Reapply stashed changes
<code>git diff</code>	Show differences between commits or files
<code>git tag <tag-name></code>	Add version tags (e.g., v1.0)
<code>git rebase <branch></code>	Reapply commits on top of another branch

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git cherry-pick  
<commit-id>
```

Apply a specific commit from another branch

GitHub Collaboration Commands

Command	Use Case
<code>git fork</code> <i>(on GitHub UI)</i>	Create your own copy of someone's repo
<code>git clone <fork-url></code>	Clone your fork
<code>git remote add upstream <original-url></code>	Connect fork to original repo
<code>git pull upstream main</code>	Sync your fork with original repo
<code>git push origin <branch></code>	Push branch to your GitHub fork

Pro Tips for Beginners

Tip	Description
<code>git log --graph --oneline --decorate</code>	See a visual tree of commits
<code>git commit --amend</code>	Edit the last commit message
<code>git reflog</code>	See all actions (useful to recover lost commits)
<code>git shortlog -sn</code>	See commit count per author (great for teamwork!)

THANK YOU!

